

Fracture Roughness Characterization from 360 Unrolled Core Images in a Sandstone Reservoir, Case study, Algeria, Hassi Messaoud

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ABSTRACT: Various existing indirect measurement methods and techniques in the petroleum industry are used to estimate some of the fracture parameters, such as sonic and electrical image logs. However, none of them can estimate the fracture roughness and the degree of mismatch between the two opposite fracture sides due to the sampling resolution. These two are important parameters to consider when dealing with geomechanics and fluid flow in tight naturally fractured reservoirs because they are important in quantifying shear strength and permeability. A quantification and detailed characterization of the joint roughness coefficient (JRC) and the degree of mismatch has been done on fractures gathered in a total of 23 wells in the Algerian Cambro-Ordovician oil-bearing formation Hassi Messaoud which consists of sandstone, clay, and some volcanic intrusions. These data have been collected from the digitalization of high-resolution unrolled 360-core photographs at the reservoir level. The results have been correlated with rock type, grain size, and fracture dip angle. Fracture roughness is ranging between 5 (low) to 19 (high) on the JRC scale and can be predicted from analysis of the general reservoir rock types (sandstone, eruptive, and clay). The results are applicable to other, especially siliciclastic reservoir lithologies, and will therefore contribute to defining a more accurate aperture and permeability estimation in building reliable Discrete Fracture Network (DFN) models.

1. INTRODUCTION

Fracture roughness is a key parameter in characterizing the permeability and fluid flow in petroleum (Camac et al., 2006) and geothermal systems (Porlles & Jabbari, 2022). As the majority of currently used large-scale models still significantly rely on simplified smooth parallel-plate models and its related models for naturally fractured reservoirs with rough walls, it is crucial to quantify the roughness in order to enhance the performance of these models (Bodin et al., 2007).

Over the past few years, there has been a growing interest in the use of core images to examine the fracture system in unconventional reservoirs, as it offers a more comprehensive and detailed representation of the fracture network (Alzubaidi et al., 2022).

In this study, the roughness of fractures and the discrepancy between fracture walls will be quantified and correlated with the general rock properties. The fracture roughness and the mismatch are critical properties to quantify rock's shear strength and hydraulic conductivity (B. Li et al., 2020). They are studied in different geoenvironmental fields such as hydrology (BAKELLI et al., 2019; Scesi & Gattinoni, 2007), and petroleum engineering (Zhao et al., 2022). Most roughness measurements come from numerical simulations (Javadi

et al., 2010), laboratory experiments (Chen et al., 2019), and near-surface rocks (B. Li et al., 2020), while measurements of deep reservoir fractures are very rare.

In this paper, we present a statistical analysis of the roughness and wall mismatch of the Cambro-Ordovician oil-bearing fractured sandstone reservoir, situated in Algeria, using data from the directional core images. The quantification of the fracture roughness has been performed using FracRough (Al-Fahmi et al., 2021), a computer software developed for the estimation of roughness and implements 360-core images.

The methodology employed in this investigation involves the use of image processing techniques to extract fracture roughness parameters from the unrolled core images. The findings of this study will contribute in enhancing comprehension of the fracture roughness in sandstone reservoirs and its influence on fluid flow, helping to optimize production and improve recovery rates in this mature oil and gas field.

2. GEOLOGICAL BACKGROUND

The study area is located in the southwestern part of the Hassi Messaoud field, northeast of the Saharan platform. Hassi-Messaoud field represents one of the most complex fields in the world and the Cambro-Ordovician sandstones are one of the main oil and gas reservoirs in

Algeria. The field is situated 353 km from the Tunisian border and 655 km southeast of Algiers (Fig.1). Geologically, it is limited to the West by the Oued Mya depression, to the South by the mole of Amguid El Biod, to the North by the Djammâa-Touggourt structure, and to the East by the depression of Ghadamès.

During geological history, this field has undergone an intense tectonic evolution characterized by distinctive compressive phases and the diagenetic transformation in the reservoir. During its burial in geological time, shaping the deposits in the present configuration. Hassi Messaoud field is presented as a vast flattened anticlinal dome, with a general north-east-south-west direction. The features affecting the reservoir are NNE-SSW sub-meridian faults as well as faults that are perpendicular to them in a NW-SE direction. This demonstrates the tectonic character of the Horst and Grabens (Sonatrach, 2007).

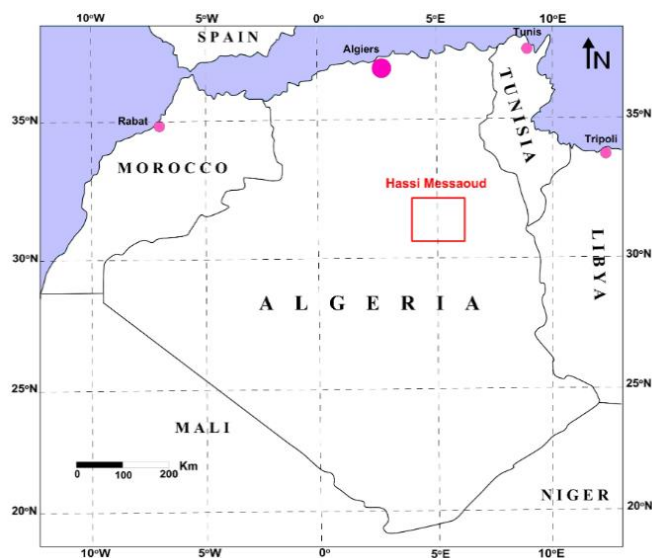


Fig. 1. Location of the study area of study. (Ameur-Zaimeche et al., 2022)

This complex and polyphased basin has been impacted, throughout the Paleozoic, by an alternation of periods of quiescence and tectonic activity. It derived a reactivation of ancient N–S, NE–SW, or NW–SE structures in arch and basin configurations (Badalini+ et al., 2002; Boudjema, 1987). This depression filled with Paleozoic deposits is northerly dipping (Logan & Duddy, 1998). The North-South structural trend in the basin is controlled by sub-vertical alignments of the Precambrian basement. Cambrian to Carboniferous strata is folded and faulted as a result of the Hercynian/Variscan compressional deformation, with major N–S and NW–SE strike orientations (Haddoum et al., 2001). The NW-SE direction of the asymmetrical folds, linked to extensive faults mainly, is generally parallel to the direction of the Ougarta Range further North (Fig.2).

3. STRATIGRAPHY

The Cambro-Ordovician sandstones are one of the most important oil reserves in Algeria. They are a large braided

fluvial system and the reservoir is divided into four lithozones, (R3, R2, Ra, Ri), based on grain size and wireline signatures (Aifa et al., 2014).

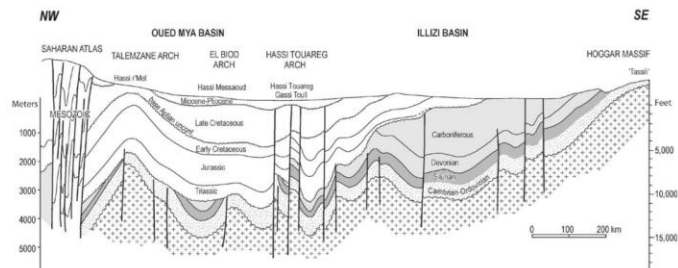


Fig. 2. Generalized structural cross-section across the Oued Mya and Illizi basins, (Guiraud et al)

Cambro-Ordovician deposits are typical of large intracratonic basins with tectonic, generally slow subsidence rates and shallow accommodation (Cherana et al., 2022; Eladj et al., 2022; Sabaou et al., 2009). In the Ordovician, the most important part consists of the Hamra quartzite, which was severely eroded at the Hercynian unconformity to form the Hassi Messaoud Ring. The Cambro-Ordovician reservoirs of the Hassi Messaoud area comprise quartzitic sandstones, which rest unconformably on granitic basement and are capped by the Hercynian unconformity. These sandstones are mature quartz arenites that are typical of those found in intracratonic basin deposition.

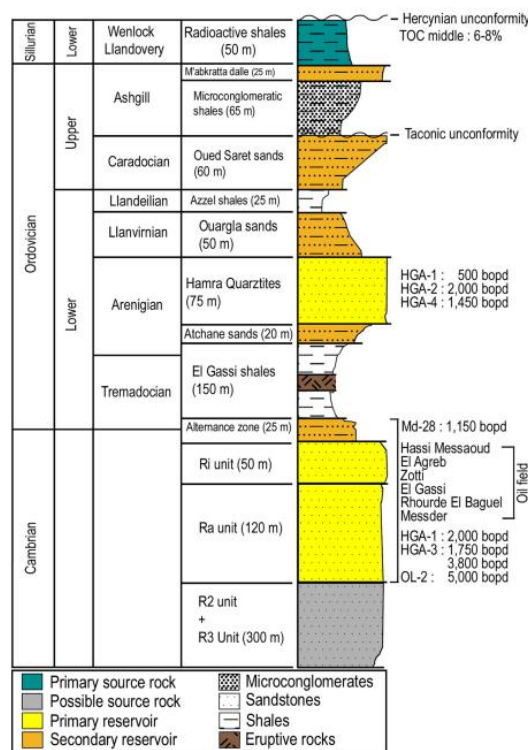


Fig.3. Lithostratigraphic column of the Cambro-Ordovician formations in Hassi Messaoud oil field area (Zerroug et al., 2007).

4. METHODOLOGY

Fracture roughness has been a well-studied subject in the field of rock mechanics for decades. There have been several methods used to describe roughness, including empirical, statistical, and fractal approaches (Tatone & Grasselli, 2010). The joint roughness coefficients (JRCs) method was widely used for 2D roughness quantification (Tse & Cruden, 1979) (and it ranges from 0 (very smooth) to 20 (very rough) (Fig.4).

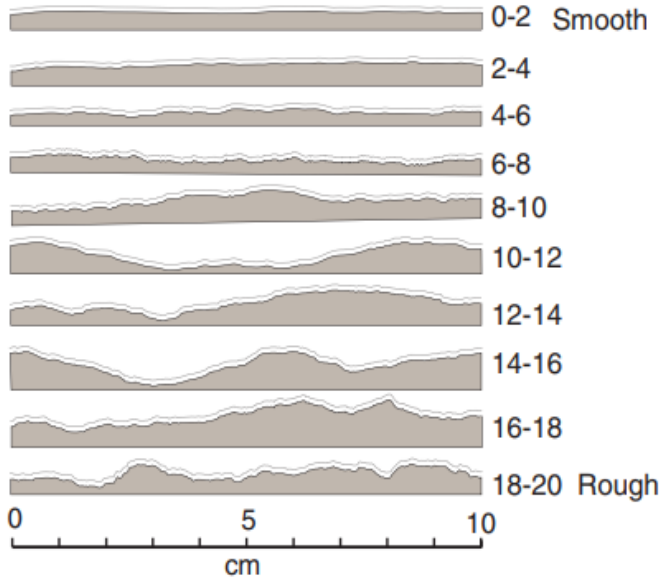


Fig. 4 Typical profiles of joint roughness coefficients (Al-Fahmi et al., 2018)

The JRC values were quantified using the following equation (Y. Li & Zhang, 2015)

$$JRC = 55.7376 Z_2 - 4.1166$$

With:

$$Z_2 = \left[\frac{1}{N_t} \sum \left(\frac{Z_{i-1} - Z_i}{x_{i-1} - x_i} \right)^2 \right]^{1/2}$$

Z_2 is defined as the root mean square of the first deviation of the 2D digitized profile.

The open and closed fractures, fracture fill, and density of fractures were all shown by the core pictures. A total of 26 vertical wells have been used in this study, with core photos available for the Cambro-Ordovician reservoir. An example of core photos is shown in Figure 5. The cores were scanned with a high-resolution scanner with a calibration of 10 pixel/mm. We have only considered natural open fractures in this study, and the core photographs were loaded and manipulated to allow the selection of natural fractures. The digitization has been done using ImageJ, an image processing and analysis software (Fig.6) and the distinction between natural fractures and induced fractures (fractures that occurs during drilling, coring, or core handling) was achieved in a visual way by looking at the dip change.

The profile of a plane fracture is a perfect sinusoid. However, actual fractures have rough and irregular upper and lower traces (Al-Fahmi et al., 2021).

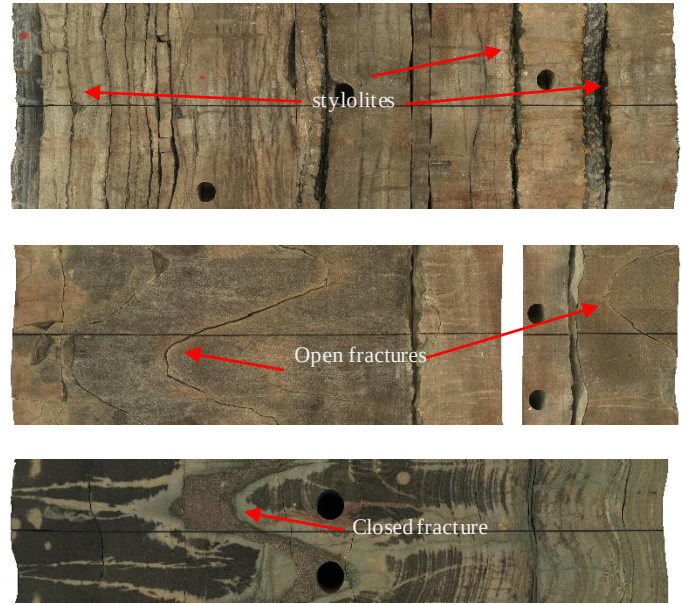


Fig. 5 photographs of the unwrapped 360° core photos (length is 1m each).

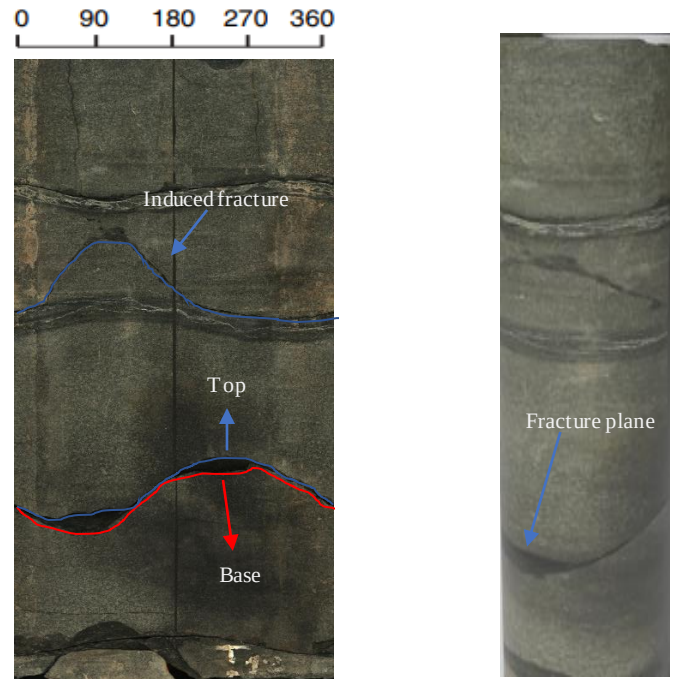


Fig. 6 examples of the 360 core photos used to digitize the fracture (length is 1m each).

The fractures characterization, roughness, dip and wall mismatch has been done using a code (FracRough) developed by (Al-Fahmi et al., 2021).

4.1. JRC correction

The parameters for JRC estimation from the fracture surface are usually two-dimensional (2D), in which case one or more straight profiles derived from the surface cannot represent the entire surface roughness. Therefore,

it is necessary to correct the apparent JRC values for dip angle. The correlation happens because higher the fracture angle, the rougher the fractures look (Fig.7a-b). The scatterplots below display the JRC and fracture dip of the 80 natural fractures that were considered in this study. Each dot on the graph represents an individual fracture and its combination of JRC and fracture dip.

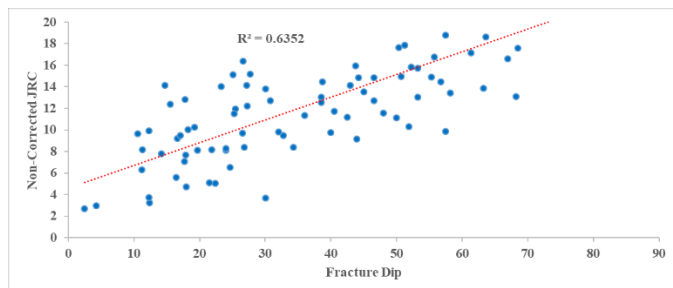


Fig. 7a. Scatterplot and linear regression between JRC and Fracture Dip.

First, the JRC values for fracture dip were regressed for 80 fractures samples, and a correlation was developed with $R = 0.6352$ (Fig. 7a). There is a correlation between JRC and fracture dip angle. As fracture dip increases, JRC also tends to increase.

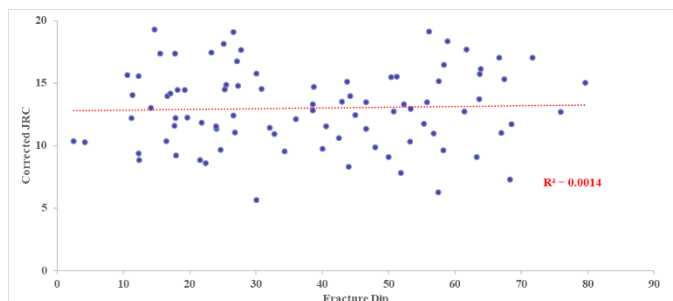


Fig. 7b Correlation JRC and Fracture Dip after correction.

A correction was then applied to the JRC values and a new correlation was developed from the same set (Fig. 7b). In this case the scatterplot showed a haphazardly behavior. The lower correlation in Fig. 7b was mainly due to the effect of the apparent dip on the JRC numbers.

5. RESULTS AND DISCUSSION

The fracture roughness distribution from this investigation are reported in the next two sub-sections (Section 5.1), and 80 JRC measurements are compared in terms of lithology (Section 5.2). Finally, we summarized the impact of roughness on aperture and we discussed the 'fabric- dependent' behavior.

5.1. Roughness Statistical Distribution

The histogram (Fig.8) showcases a normal statistical distribution for the JRC of the digitized open fractures, also the majority (88%) of the JRC values fall within the range of 8 to 18.

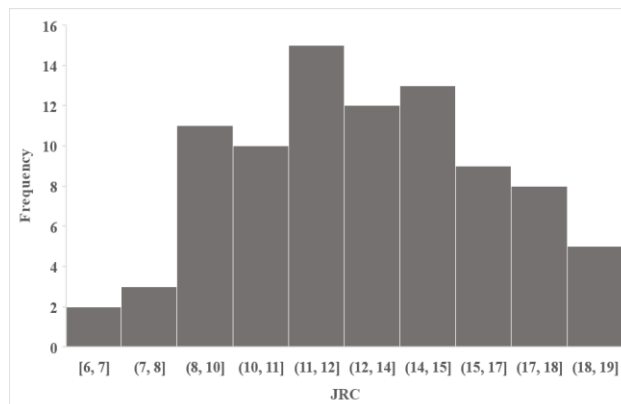


Fig. 8. JRC histogram of the fractures based on JRC values.

The data in Fig. 8 indicates that most of the JRC values are above 6, and the JRC values with the highest frequency lie between [11,12] and [14, 15]. It is shown in the frequency histogram that the least frequent JRC classes are attributed to two intervals, centered around [6, 7] and [7, 8].

5.2. Roughness and fabric dependent Behavior

In the scattergram in Fig.9, the relationship between JRC values and depth are investigated. Each data point corresponds to an individual fracture. The graph shows random behavior of the points. As the depth increases, there is no set expectation of what the JRC will do. However, a careful examination demonstrates that the fracture roughness in Hassi Messaoud can be described within three different geological realms: the Hamra Quartzite (Arenigian, Upper Ordovician), El Gassi shales (Tremadocian, Lower Ordovician), and the Cambrian sandstones.

The Ordovician shows a cloud with roughly high JRC, ranging between 8 and 18, in the depth range of 3250-3430 m. However, at depth 3430 m the JRC dramatically decreases; this phenomenon characterizes the shales. In the same context, the Cambrian exhibits a clear positive JRC trend, as the depth increases the JRC tend to increase.

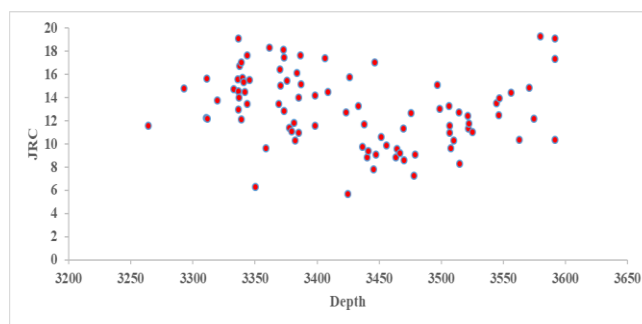


Fig. 9. Scatterplot of Depth in function of JRC. The vertical blue lines serve as a threshold and delimitate the three

In a second interpretation, we tried to better understand the relationship between depth, lithology, and JRC. The roughness and mismatch of the fracture walls reflect the

lithology of the host rock and therefore are considered “fabric-dependent” (Al-Fahmi et al., 2018). In this respect, we attempted to narrow our study by embedding the lithology in our analyses. The scatterplot displayed in fig. 10 shows a decreasing-increasing trend. The lower points figure at the depth range of 3450-3500 m. Based on the comparison with the 360 enrolled cores, we associate this trend with the magmatic rocks that intrude the Cambro-Ordovician formation.

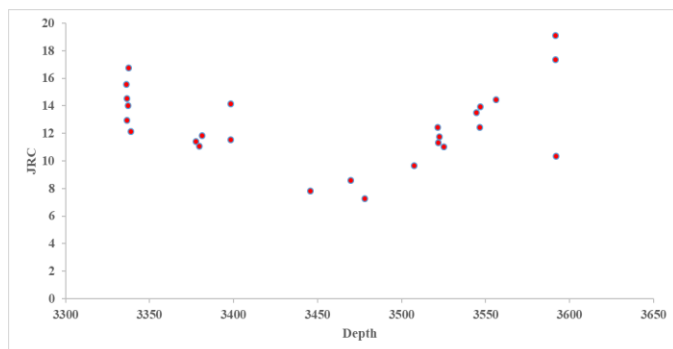


Fig. 10. Scatterplot of Depth in function of JRC of magmatic rocks.

Another classification based on the grain size (Fig. 11) has been adopted to better understand the roughness distribution in the Hassi Messaoud sandstone field. We observed that the very fine grain sandstone (VF) occurs at depth 3390 m, corresponding to the upper limit of the El-Gassi shales. The JRC seems to follow a vertical line with ranges between 10 and 16. In the opposite side, the coarse (C) grains sandstone appears in the Cambrian with JRCs not exceeding 13. Finally, medium (M) and fine (F) grained sandstones have a random distribution along the formations.

Texture (grain size)		Sediment Name	Rock Name
Coarse (over 2 mm)		Gravel (rounded fragments)	Conglomerate
		Gravel (angular fragments)	Breccia
Medium (1/16 to 2 mm)		Sand	Sandstone
Fine (1/16 to 1/256 mm)		Silt	Siltstone
Very Fine (less than 1/256)		Clay (also known as Mud)	Shale (if stratified) Mudstone (if non-stratified)

Fig. 10. Classification of sandstones based on grain size.

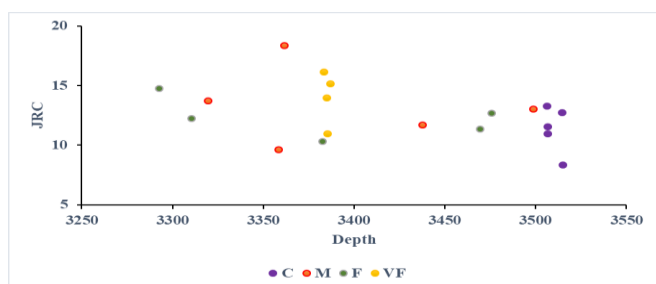


Fig. 11. Scatterplot of Depth in function of JRC of Sandstones.

The distribution of clays ranges between 3340 and 3370 m and corresponds to the El-Gassi shales. The JRC ranges between 4 and 14, not noticeable in all depths, but present enough around 3450 ft.

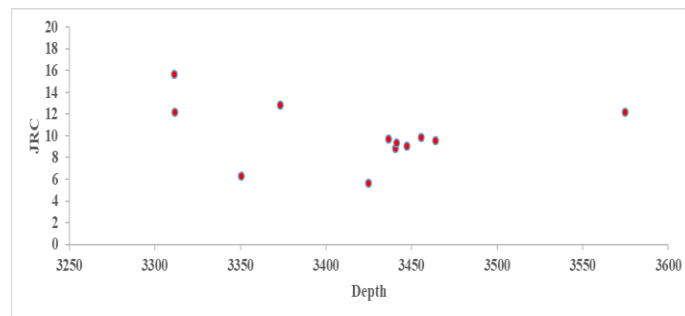


Fig. 12. Scatterplot of Depth in function of JRC of Sandstones

5.3. Effect of Fracture Roughness on Aperture

We investigated the relationship between JRC number and aperture, two important parameters that control fluid flow in joints and natural fractures. The 2D aperture in each image was measured by determining the distance between the upper and lower fracture wall at evenly spaced spots. When considering joint and fracture shearing, roughness affects the change in aperture distribution. Thus, the distributions of contact area and void space determine the fracture dilation and hydraulic properties during shearing (Liu et al., 2015). Matching and mismatching quantification provides insight into what relative values of shear stresses can be expected for different roughness profiles and discontinuity surfaces. Mismatch coefficient is represented by numbers between 0 and 1 (fig. 13b). The JRC values for the 80 natural fracture aperture distributions was plotted against the aperture standard deviation of the fractures (Fig. 13a). Predictably, the matched surfaces of fractures that have 0 to 0.1 standard deviation express the largest top-base correlation, or contact area, compared to fractures that display an aperture of more than 0.15. Consequently, these samples exhibit a top-base correlation of less than 0.85 and correspond to the fractures affected by shearing (Fig. 13a, b, and c).

6. CONCLUSION

In this study, the 2D fracture roughness and mismatch have been quantified and analyzed. A total of 80 fractures were obtained from the digitizing of unrolled core images acquired from 26 cored wells in Hassi Messaoud field, Algeria. The following conclusions can be obtained:

- The Cambro-Ordovician formation in Hassi Messaoud exhibits medium to high roughness between 5 to 19 on the Joint Roughness Coefficient

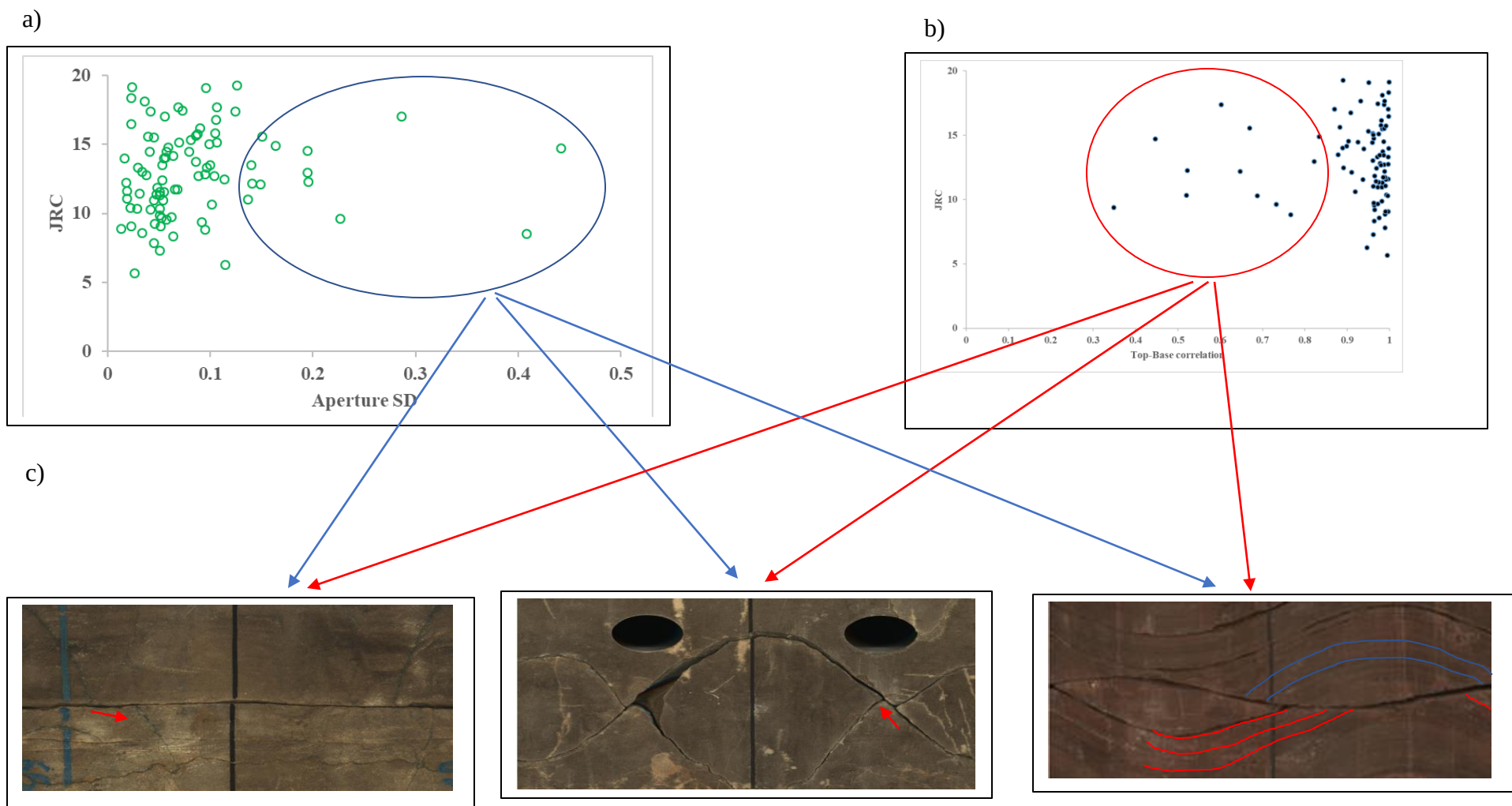


Fig13: a) Standard deviation of the aperture b) Top base correlation (mismatch coefficient) c) Examples of shear Fractures

- Fracture roughness associated with intrusive rocks (plutonic rocks) decreases with depth until it reaches El Gassi shales and then the fracture roughness gradually increases with depth. This may be due to the tectonic events that happened after the intrusion.
- All the fractures that showed a mismatch are shear fractures
- The matched rough fracture produces noticeably smaller and almost similar apertures. The mismatched rough fractures are characterized by very large apertures.

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REFERENCES

1. Al\'vifa, T., Zerrouki, A. A., Baddari, K., & Géraud, Y. (2014). Magnetic susceptibility and its relation with fractures and petrophysical parameters in the tight sand oil reservoir of Hamra quartzites, southwest of the Hassi Messaoud oil field, Algeria. *Journal of Petroleum Science and Engineering*, 123, 120–137.
2. Al-Fahmi, M. M., Ozkaya, S. I., & Cartwright, J. A. (2018). New insights on fracture roughness and wall mismatch in carbonate reservoir rocks. *Geosphere*, 14(4), 1851–1859.
3. Al-Fahmi, M. M., Ozkaya, S. I., & Cartwright, J. A. (2021). FracRough—Computer program to calculate fracture roughness from reservoir rock core. *Applied Computing and Geosciences*, 9, 100045.
4. Alzubaidi, F., Makuluni, P., Clark, S. R., Lie, J. E., Mostaghimi, P., & Armstrong, R. T. (2022). Automatic fracture detection and characterization from unwrapped drill-core images using mask R-CNN. *Journal of Petroleum Science and Engineering*, 208, 109471.
5. Ameer-Zaimeche, O., Kechiched, R., Heddam, S., & Wood, D. A. (2022). Real-time porosity prediction using gas-while-drilling data and machine learning with reservoir associated gas: Case study for Hassi Messaoud field, Algeria. *Marine and Petroleum Geology*, 140, 105631.
6. Badalini, G., Redfern, J., & Carr, I. D. (2002). A synthesis of current understanding of the structural evolution of North Africa. *Journal of Petroleum Geology*, 25(3), 249–258.
7. BAKELLI, O., CHALLALI, M., & ZEDDOURI, A. (n.d.). *Hydrogeological study of the lower Devonian aquifer in The Illizi region*.
8. Bodin, A., Bäckdahl, H., Fink, H., Gustafsson, L., Risberg, B., & Gatenholm, P. (2007). Influence of cultivation conditions on mechanical and morphological properties of bacterial cellulose tubes. *Biotechnology and Bioengineering*, 97(2), 425–434.
9. Boudjema, A. (1987). *Évolution structurale du bassin pétrolier triasique du Sahara Nord oriental (Algérie)*. Paris 11.
10. Camac, B. A., Hunt, S. P., & Boulton, P. J. (2006). Local rotations in borehole breakouts—observed and modeled stress field rotations and their implications for the petroleum industry. *International Journal of Geomechanics*, 6(6), 399–410.
11. Chen, Y., Lian, H., Liang, W., Yang, J., Nguyen, V. P., & Bordas, S. P. A. (2019). The influence of fracture geometry variation on non-Darcy flow in fractures under confining stresses. *International Journal of Rock Mechanics and Mining Sciences*, 113, 59–71.
12. Cherana, A., Aliouane, L., Doghmane, M. Z., Oudfeul, S.-A., & Nabawy, B. S. (2022). Lithofacies discrimination of the Ordovician unconventional gas-bearing tight sandstone reservoirs using a subtractive fuzzy clustering algorithm applied on the well log data: Illizi Basin, the Algerian Sahara. *Journal of African Earth Sciences*, 196, 104732.
13. Eladj, S., Lounissi, T. K., Doghmane, M. Z., & Djeddi, M. (2022). Wellbore Stability Analysis Based on 3D Geo-Mechanical Model of an Algerian Southeastern Field. *Advances in Geophysics, Tectonics and Petroleum Geosciences: Proceedings of the 2nd Springer Conference of the Arabian Journal of Geosciences (CAJG-2), Tunisia 2019*, 615–618.
14. Haddoum, H., Guiraud, R., & Moussine-Pouchkine, A. (2001). Hercynian compressional deformations of the Ahnet–Mouydir Basin, Algerian Saharan Platform: far-field stress effects of the Late Palaeozoic orogeny. *Terra Nova*, 13(3), 220–226.
15. Javadi, M., Sharifzadeh, M., & Shahriar, K. (2010). A new geometrical model for non-linear fluid flow through rough fractures. *Journal of Hydrology*, 389(1–2), 18–30.
16. Li, B., Ye, X., Dou, Z., Zhao, Z., Li, Y., & Yang, Q. (2020). Shear strength of rock fractures under dry, surface wet and saturated conditions. *Rock Mechanics and Rock Engineering*, 53, 2605–2622.
17. Li, Y., & Zhang, Y. (2015). Quantitative estimation of joint roughness coefficient using statistical parameters. *International Journal of Rock Mechanics and Mining Sciences*, 77, 27–35.
18. Liu, R., Jiang, Y., Li, B., & Wang, X. (2015). A fractal model for characterizing fluid flow in fractured rock masses based on randomly distributed rock fracture networks. *Computers and Geotechnics*, 65, 45–55.
19. Logan, P., & Duddy, I. (1998). An investigation of the thermal history of the Ahnet and Reggane Basins, Central Algeria, and the consequences for hydrocarbon generation and accumulation. *Geological Society, London, Special Publications*, 132(1), 131–155.
20. Porlles, J. W., & Jabbari, H. (2022). Simulation-Based Patterns Optimization of Enhanced Geothermal Systems. *56th US Rock Mechanics/Geomechanics Symposium*.
21. Sabaou, N., Ait-Salem, H., & Zazoun, R. S. (2009). Chemostratigraphy, tectonic setting and provenance of the Cambro-Ordovician clastic deposits of the subsurface Algerian Sahara. *Journal of African Earth Sciences*, 55(3–4), 158–174.

22. Scesi, L., & Gattinoni, P. (2007). Roughness control on hydraulic conductivity in fractured rocks. *Hydrogeology Journal*, 15, 201–211.
23. Tatone, B. S. A., & Grasselli, G. (2010). A new 2D discontinuity roughness parameter and its correlation with JRC. *International Journal of Rock Mechanics and Mining Sciences*, 47(8), 1391–1400.
24. Tse, R., & Cruden, D. M. (1979). Estimating joint roughness coefficients. *International Journal of Rock Mechanics and Mining Sciences & Geomechanics Abstracts*, 16(5), 303–307.
25. Zerroug, S., Ali Sahraoui, F., & Bouarissa, N. (2007). Structural parameters and pressure coefficients for CdS x Te 1-x: FP-LAPW calculations. *The European Physical Journal B*, 57, 9–14.
26. Zhao, Z., Wang, S., Wang, Z., Wang, S., Ma, C., & Yang, B. (2022). Surface roughness stabilization method based on digital twin-driven machining parameters self-adaption adjustment: A case study in five-axis machining. *Journal of Intelligent Manufacturing*, 1–10.